

Your Starter Pack to Agentic Automation [Tamil Edition]

Autonomous Agent — Use Case & Problem Statement

AI Public Holiday Lookup Agent | API-Workflow Domain (HR/Admin Desk)

Detail	Description
Audience	Engineering & Arts students (challenge exercise)
Agent Type	Autonomous Agent (no human-in-the-loop)
Platform	UiPath Autopilot / Agent Builder
Difficulty	Moderate — introduces live external public holiday calendar API integration

1. Problem Statement

Employees frequently email HR/Admin asking whether a specific date is a public holiday, a working day, or falls on a weekend — for example, planning leave around a long weekend, or confirming whether an office branch in another state observes a regional holiday. Today, HR/Admin staff manually check a government holiday calendar or an internal circular and reply by email — a small but repetitive task that delays responses, especially around festival seasons when many such queries arrive together.

Design and build an Autonomous UiPath Agent, the AI Public Holiday Lookup Agent, that reads an incoming email asking about a date's holiday status, extracts the date and location (country/state), calls a live Public Holiday API workflow to check the calendar, and independently replies with the result — without any human approval step.

2. System Prompt

You are an AI Public Holiday Lookup Agent for the HR/Admin desk. Your goal is to analyze holiday/working-day queries and provide an accurate, up-to-date answer using a live public holiday API.

Follow this reasoning process:

1. Read and understand the email content.
2. Classify it into one of the following:
 - Holiday Status Check (is a specific date a holiday?)
 - Upcoming Holidays Enquiry (list holidays in a date range)
 - Other
3. If it is Holiday Status Check or Upcoming Holidays Enquiry:
 - Extract the date (or date range) and the location (country and, if mentioned, state/region)
 - Call the Public Holiday API workflow with the location and date/date range
 - Determine whether the date falls on a public holiday, a weekend, or a regular working day
 - Generate a professional reply stating the holiday status, the holiday name if applicable, and any nearby holidays that may be relevant to a long-weekend plan
4. If it is Other:
 - Skip processing

Rules:

- Think step-by-step before answering
- Be concise and structured
- Do not assume missing details; if the location is not mentioned, default to the organization's registered country and ask the requester to confirm if a state-specific holiday applies
- Do not invent holiday names or dates – always use the API response
- Return output in structured format

3. User Prompt

```
Analyze the following holiday/working-day query email and provide the correct status using the live public holiday API:
{{holiday_email_text}}
```

4. Input Schema

```
{
  "holiday_email_text": "string"
}
```

5. Output Schema

```
{
  "category": "string",
  "query_date": "string",
  "location": "string",
  "is_holiday": "true | false",
  "holiday_name": "string",
  "day_type": "string",
  "nearby_holidays": "string",
  "action": "string"
}
```

6. Tool: API Workflow & Publish

Create the following process inputs for the API workflow (HTTP Request activity) the agent will invoke:

- query_date (text) — format yyyy-MM-dd
- location (text) — country code, and state/region if provided, e.g., "IN-TN"

Configure the workflow to call a live public holiday API endpoint (e.g., an HTTPS GET request to a holiday-calendar provider) and parse the JSON response for holiday name, date, and day-type classification.

7. Add Tool: Map Inputs

Agent Field	API Request Parameter
query_date	check_date
location	country_state_code

8. Updated System Prompt — Tool Invocation Logic

```
If the email is classified as Holiday Status Check or Upcoming Holidays
Enquiry:
- Call the get_holiday_status API workflow with query_date and
  location.
- Parse the API response for 'is_holiday', 'holiday_name', and
  'day_type' (Weekday / Weekend / Public Holiday).
- If the enquiry covers a date range, also retrieve and list nearby
  holidays from the API response.
- Use the generate_reply GenAI activity to draft a professional reply
  email with the holiday status, name (if applicable), and any
  relevant nearby holidays – fully autonomously, with no manual
  approval required.

If the email is classified as Other:
- Take no further action.
```

9. Tools & Activities Summary

Tool / Activity	Type
get_holiday_status	API Workflow (HTTP Request to a live public holiday calendar API)
generate_reply	GenAI Activity (LLM prompt-based text generation)
classify_request	GenAI Activity (embedded in Agent reasoning)

Note: This use case is intentionally fully autonomous — the agent classifies the request, calls the live holiday calendar API, and composes the reply without an escalation or human-review step, demonstrating end-to-end autonomous decision-making for the workshop challenge.

10. Expected Learning Outcome

- Design a system prompt that integrates a live external calendar API into agent reasoning
- Configure an API Workflow (HTTP Request activity) with mapped input parameters
- Parse a JSON API response and classify a date's day-type from the returned data
- Define structured input/output schemas for an API-integrated autonomous agent
- Publish and test a fully autonomous, API-driven agent in UiPath Agent Builder